Md Rezwanul **Haque**

MASc Candidate, University of Waterloo (UW), Canada | ML Scientist & Engineer

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*My career spans top academic institutions, including the **University of Waterloo (UW), Canada (CA)** and **Khulna University of Engineering & Technology (KUET), Bangladesh (BD)**. As a machine learning specialist, I have collaborated with multidisciplinary teams in academia and industry, contributing to impactful research and real-world AI solutions. I possess a strong publication record in top-tier venues and have been recognized for my contributions to data science and applied machine learning. My research focuses on Agentic Systems for Embodied AI, including Multi-Agent Embodied Systems with Shared Visual Understanding, as well as World Models and Visual Cognition for Embodied Agents.*

EDUCATION

University of Waterloo — ***Waterloo, CA***

PhD student in Electrical and Computer Engineering (ECE) — *May 2026 – p.*

- **Supervisor(s):** *Prof. Fakhri Karray & Prof. Zhou Wang*
- **Research Focus:** Agentic Systems for Embodied AI, Multi-Agent Embodied Systems with Shared Visual Understanding, and World Models and Visual Cognition for Embodied Agents.

University of Waterloo — ***Waterloo, CA***

MASc in Electrical and Computer Engineering (ECE) — *May 2024 – April 2026.*

- **Thesis:** "Audio-Visual Feature Fusion through Transformers for Automated Depression Screening in Social Media Content."
- **Supervisor(s):** *Prof. Fakhri Karray & Prof. Pin-Han Ho*

Khulna University of Engineering & Technology (KUET) — ***Khulna, BD***

Bachelor of Science in Computer Science and Engineering (CSE) — *Dec. 2014 - Feb. 2019*

- **Thesis:** "A Study on Non-Invasive Hemoglobin Measurement Techniques and Predictions."
- **Supervisor(s):** *Prof. Dr. M.M.A Hashem*

EMPLOYMENT

University of Waterloo — ***Waterloo, CA***

Graduate Research Assistant — *May 2024 – p.*

- Centre for Pattern Analysis and Machine Intelligence (CPAMI) Lab

University of Waterloo — ***Waterloo, CA***

Teaching Assistant — *Sept. 2024 - p.*

- ECE 222: Digital Computers (Fall 2025) – Students: 300
- ECE 459: Programming for Performance (Winter 2025, Winter 2026) – Students: 453, 440
- ECE 252: Systems Programming and Concurrency (Fall 2024, Spring 2025) – Students: 206, 210

Apsis Solutions Ltd. — ***Dhaka, BD***

Machine Learning Engineer — *Mar. 2021 - Apr. 2024*

- Developed and implemented models for Bangla Handwritten and Printed OCR.
- Engineered and deployed requirement-based OCR APIs for various applications.
- Worked on Quality Assurance (QA) and Text Normalization for a Virtual Personal Assistant (VPA).

EBLICT-Dream71 Bangladesh Ltd. — ***Dhaka, BD***

Machine Learning Consultant — *Jan. 2024 - Apr. 2024*

- Provided expertise and development support for a Bangla Virtual Private Assistant.

Bengali.AI — ***Dhaka, BD***

Computer Vision and NLP Researcher — *Jan. 2021 - Apr. 2024*

- Conducted research and development for Bengali OCR on both handwritten and printed text.
- Worked on core NLP tasks including Grapheme Parsing, Unicode Normalization, and building a Bengali Common Voice Speech dataset.
- Developed models for Document Layout Analysis (DLA) to understand document structures.

Independent University, Bangladesh (IUB) ***Dhaka, BD***

PART-TIME RESEARCH ENGINEER *Jan. 2022 - Nov. 2022*

- In the Electrical and Electronic Engineering (EEE) department, conducted research on designing a highly efficient, non-invasive device for measuring human blood component levels using Genetic Algorithms and DNN-based models.

PUBLICATIONS

*My work on multimodal, explainable, and reliable AI has been presented at top-tier conferences (**CVPR, ICCV, IJCNN, SMC,** etc.) and published in leading journals (**IEEE Access, Elsevier, Springer,** etc.). I have made contributions to substantial research on computer vision, biological diagnostics, and human–AI interaction; a number of my publications have received remarkable recognition and citations. (Google Scholar)*

SPOTLIGHT ( ) ORAL ( ) ≥ 100 CITATIONS ( ) BEST PAPER ( ) BEST POSTER AWARD ( ) PATENT ( )
BOOK CHAPTER ( ) EQUAL CONTRIBUTION (*) IMPACT FACTOR (I.F.) IMPACT SCORE (I.S.) RANK ( )

H-INDEX	i10-Index	MAX CITATIONS	TOTAL CITATIONS
10	10	451 [J_5]	**1309**

3	2	5	0	1	0	0

Journal Proceedings

J_1 – **2021** *IEEE* I.F. 3.9 (Q1)
Md. Rezwanul Haque, S. M. Taslim Uddin Raju, Md. Asaf-uddowla Golap, M.M.A Hashem, *"A Novel Technique for Non-Invasive Measurement of Human Blood Component Levels from Fingertip Video Using DNN Based Models"*, IEEE Access, IEEE, vol. 9, pp. 19025–19042, Jan. 2021.

J_2 – **2021** *Elsevier* I.F. 4.9 (Q1)
M. A.-u. Golap, S.T.U. Raju, **Md. Rezwanul Haque**, M.M.A Hashem, *"Hemoglobin and Glucose Level Estimation from PPG Characteristics Features of Fingertip Video Using MGGP-Based Model"*, Biomedical Signal Processing and Control, Elsevier. vol. 67, pp. 102478, Mar. 2021.

J_3 – **2021** *Springer* I.S. 2.4 (Q2)
S. M. A. Ullah, Md. Milon Islam, Saifuddin Mahmud, Sheikh Nooruddin, S.M. Taslim Uddin Raju, **Md. Rezwanul Haque**, *"Scalable Telehealth Services to Combat Novel Coronavirus (COVID-19) Pandemic"*, SN Computer Science, Springer Nature, vol. 2, no. 1, pp. 18, Jan. 2021.

J_4 ★ **2021** *Springer* I.S. 2.4 (Q2)
Laboni Akter, Ferdib-Al-Islam, Md. Milon Islam, Mabrook S. Al-Rakhami, **Md. Rezwanul Haque** , *"Prediction of Cervical Cancer from Behavior Risk Using Machine Learning Techniques"*, SN Computer Science, Springer Nature, vol. 2, no. 3, pp. 177, Mar. 2021.

J_5 ★ **2020** *Springer* I.S. 2.4 (Q2)
Md. Milon Islam, **Md Rezwanul Haque**, Hasib Iqbal, Md. Munirul Hasan, Mahmudul Hasan, Muhammad Nomani Kabir, *"Breast Cancer Prediction: A Comparative Study Using Machine Learning Techniques"*, SN Computer Science, Springer Nature, vol. 1, no. 5, pp. 290, Sep. 2020.

J_6 ★ **2020** *Springer* I.S. 2.4 (Q2)
Amanullah Asraf, Md. Zabirul Islam, **Md. Rezwanul Haque**, Md. Milon Islam, *"Deep Learning Applications to Combat Novel Coronavirus (COVID-19) Pandemic"*, SN Computer Science, Springer Nature, vol. 1, no. 6, pp. 363, Nov. 2020.

J_7 – **2025** *IEEE* I.F. 9.8 (Q1)
S M Taslim Uddin Raju, **Md Rezwanul Haque**, Md. Milon Islam, Hamdi Altaheri, and Fakhri Karray, *"ClusGNN-ViT: Integrating Deep Embedded Clustering and GNN-MIL for Microscopic Image Captioning and Diagnosis"*, Expert Systems With Applications (Elsevier), 2026. [Under Review]

Conference Proceedings

C_1 **2026** *SMC* B
Md Rezwanul Haque, Md. Milon Islam, and Fakhri Karray, *"Towards Robust Reinforcement Learning for Small-Scale Language Model Agents"*, IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2026. [Under Review]

C_2 – **2026** *Photonics North* –
M. F. A. Fattah, H. Abouali, S. Srikant, **Md Rezwanul Haque**, N. Barra, Fakhri Karray, J. Schertzer, M. Poudineh, and D. Ban, *"Quantum Dot-Encoded Time Series Signals Decoded with Machine Learning Algorithms for Continuous Detection of Multiple Biomarkers"*, Photonics North, Quebec City Convention Centre, Canada, 2026. [Abstract Accepted]

C_3		**2025**	*SMC*	B	**Md Rezwanul Haque**, Md. Milon Islam, S M Taslim Uddin Raju, Hamdi Altaheri, Lobna Nassar, and Fakhri Karray, *"MDD-Net: Multimodal Depression Detection through Mutual Transformer"*, IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2025.
C_4		**2025**	*SMC*	B	**Md Rezwanul Haque**, Md. Milon Islam, S M Taslim Uddin Raju, Hamdi Altaheri, Lobna Nassar, and Fakhri Karray, *"MMFformer: Multimodal Fusion Transformer Network for Depression Detection"*, IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2025.
C_5	–	**2025**	*IJCNN*	A	S.T.U Raju, Md. Milon Islam, **Md Rezwanul Haque**, Hamdi Altaheri, and Fakhri Karray, *"GNN-ViTCap: GNN-Enhanced Multiple Instance Learning with Vision Transformers for Whole Slide Image Classification and Captioning"*, International Joint Conference on Neural Networks (IJCNN), 2025.
C_6	–	**2023**	*ATC*	C	M. Rumi, B. Ullah, **Md. Rezwanul Haque**, Abdullah Al Noman, Emranul Haque, and Feroz Ahmed, *"Smartphone Based BP Level Monitoring System Using DNN Model"*, In 2023 International Conference on Advanced Technologies for Communications (ATC), pp. 12–18, 19-21 Oct. 2023.
C_7	–	**2023**	*ICCIT*	C	M. Rumi, B. Ullah, **Md. Rezwanul Haque**, A.A. Noman, E. Haque, and Dr. Feroz Ahmed, *"Body Weight Estimation Using Smartphone Based Photoplethysmography Signal"*, 26th International Conference on Computer and Information Technology (ICCIT), pp. 1-6, 13-15 Dec. 2023.
C_8	–	**2023**	*ICDAR*	A	**Md Rezwanul Haque** et al., *"BaDLAD: A Large Multi-Domain Bengali Document Layout Analysis Dataset"*, ICDAR 2023: Document Analysis and Recognition - ICDAR, pp. 326–341, 19 Aug. 2023.
C_9	★	**2018**	*IC4ME2*	–	**Md. Rezwanul Haque**, Md. Milon Islam, H. Iqbal, M.S. Reza, M. K. Hasan, *"Performance Evaluation of Random Forests and Artificial Neural Networks for the Classification of Liver Disorder"*, Proc. International Conference on Computer, Communication, Chemical, Material and Electronic Engineering (IC4ME2), IEEE, pp. 1-5, 8-9 Feb. 2018.
C_{10}	★	**2017**	*R10HTC*	–	Md. Milon Islam, H. Iqbal, **Md. Rezwanul Haque**, M.S. Reza, M. K. Hasan, *"Prediction of breast cancer using support vector machine and K-Nearest neighbors"*, Proc. 2017 IEEE region 10 humanitarian technology conference (R10-HTC), IEEE, pp. 226–229, 21–23 December 2017.

Workshop Proceedings

W_1	–	**2026**	*CVPR*	A*	MSR Shujon, MJH Jim, MH Islam, **Md Rezwanul Haque**, MR Islam, H Altaheri, Fakhri Karray, *"CAST: Channel-Aware Spatial Transfer Learning with Pseudo-Image Radar for Sign Language Recognition"*, Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2026.[Accepted]
W_2	–	**2026**	*ACL*	A*	MSR Shujon, MJH Jim, MH Islam, **Md Rezwanul Haque**, Fakhri Karray, *"The Blackwell Collective at StanceNakba Shared Task: PAST-TIDE: Prototype-Anchored Statement Tuning with Topic-Invariant Normalization for Stance Detection"*, Proceedings of the 2nd International Workshop on Nakba Narratives as Language Resources, ACL Anthology, 2026. Accepted March 21, 2026.[Accepted]
W_3		**2025**	*ICCV*	A*	**Md Rezwanul Haque***, Md. Milon Islam*, S M Taslim Uddin Raju, Fakhri Karray, *"A Signer-Invariant Conformer and Multi-Scale Fusion Transformer for Continuous Sign Language Recognition"*, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, pp. 4931-4940, 2025. [Media Coverage]
W_4		**2025**	*ICCV*	A*	Md. Milon Islam*, **Md Rezwanul Haque***, S M Taslim Uddin Raju, Fakhri Karray, *"FusionEnsemble-Net: An Attention-Based Ensemble of Spatiotemporal Networks for Multimodal Sign Language Recognition"*, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, pp. 4924-4930, 2025.

HONORS & AWARDS

Throughout my academic career, I have been consistently recognized with numerous scholarships and awards for excellence at the international, national, regional, and university levels, securing significant funding for my research and studies.

2025	Recipient	***Faculty of Engineering (FOE) Award (Spring)***	CAD $1500	*UW, CA*
2025	Recipient	***Best Poster Award***	–	*ICCV, USA*
2025	Recipient	***Winner (2nd) of MSLR-2025 Challenge***	USD $750	*ICCV, USA*
2025	Recipient	***Faculty of Engineering (FOE) Award (Winter)***	CAD $1,500	*UW, CA*
2025	Recipient	***Graduate Student Contingency Bursary***	CAD $5,000	*UW, CA*
2024–p.	Recipient	***Graduate Research Studentship (GRS)***	CAD $18,000/yr.	*UW, CA*
2023	Recipient	***IICT Machine Learning Idea Scholarship***	BDT 20,000	*KUET, BD*
2019	Recipient	***KUET Undergrad Thesis Scholarship***	BDT 10,000	*KUET, BD*
2015–18	Recipient	***KUET Technical Scholarship***	BDT 32,000	*KUET, BD*
2015	Recipient	***Higher Secondary Excellence Result Award***	–	*BD*
2013	Recipient	***Secondary Excellence Result Award***	–	*BD*
2011	Recipient	***School Level Award from Fisheries Subdivision***	BDT 1,000	*BD*
2010–11	Recipient	***High School General Grade Scholarship***	BDT 5,100	*BD*
2007–09	Recipient	***Primary School General Grade Scholarship***	BDT 3,500	*BD*

TECHNICAL PROFILE

2015–p.	Profile	***Stack Overflow***	2900+ Reputations	*Q/A*
2014–p.	Profile	***Codeforces***	350+ Problems Solved	*Programming*
2014–p.	Profile	***UVA Online Judge***	100+ Problems Solved	*Programming*
2019	Kaggle	***LANL Earthquake Prediction***	Top 16%	*Data Science*
2019	Kaggle	***Predict Future Sales***	Top 23%	*Data Science*
2018	Kaggle	***Toxic Comment Classification***	Top 32%	*Data Science*
2022	Course	***Quantum Machine Learning***	–	*edX*
2020	Course	***Deep Learning with Python and PyTorch***	–	*edX*
2018	Course	***Deep Learning Specialization***	–	*Coursera*
2017	Course	***Machine Learning***	–	*Coursera*

SCIENTIFIC REVIEWING

2025–p.	Reviewer	***International Conference on Computer Vision (ICCV)***	*I*
2025–p.	Reviewer	***NeurIPS Workshop MusIML***	*I*
2025–p.	Reviewer	***International Conference on Electrical Information and Comm. Technology***	*I*
2025–p.	Reviewer	***International Conference on Systems, Man, and Cybernetics (SMC)***	*I*
2025–p.	Reviewer	***International Joint Conference on Neural Networks (IJCNN)***	*I*
2023–p.	Reviewer	***PLOS One: A peer-reviewed open access scientific journal***	*II*

OTHER SERVICE

2025	Mentor	***Mentorship Program Fall 2025 (Grad Students)***	*UW, CA*
2018–19	Instructor	***Hardware Acceleration Club of KUET (HACK)***	*KUET, BD*
2015–19	Instructor	***Special Group Interested in Programming Contest (SGIPC)***	*KUET, BD*
2017	Volunteer	***University-Industry Collaboration: Challenges and Opportunities***	*KUET, BD*
2017	Volunteer	***3rd International Conference on Electrical Information and Communication Technology (EICT)***	*KUET, BD*
2015–17	Organizer	***National High School Programming Contest (NHSPC)***	*KUET, BD*
2014	Participant	***Regional Mathematical Olympiad 2014***	*BD*